

Computer Architectures

Exam of 12.2.2025 - part I

First name, Last name, ID.....

A - Question #2

Consider a processor connected to 64 KB of memory and equipped with a set-associative cache consisting of 16 sets of 2 lines each, for a total of 32 different lines, each of 32 bytes. Assuming that:

- The cache is initially not empty, and its configuration is reported in the Cache table.
- The adopted replacement algorithm is the Least Frequently Used (LFU). In the Cache table, blocks are marked with an increasing counter, starting at 0 when the block is cached, which indicates how many times each block has been accessed. In case the counters are equal, the oldest value, marked with an arrow, is replaced. In case a new block is inserted into the cache, the pointer to the oldest value is updated.

Given the sequence of memory accesses shown in the Accesses table, determine the corresponding set and line being accessed. Use the Cache table to help you calculate the set and the line involved in each operation. This way, you will get the final cache status. Finish by providing the total number of hits and misses that occurred.

Accesses

Block	Block (Binary)	Accessed Set	Accessed Line	H/M
367	0001 0110 1111	15	31	H
240	0000 1111 0000	0	1	M
2005	0111 1101 0101	5	10	H
2020	0111 1110 0100	4	9	H
152	0000 1001 1000	8	17	H
254	0000 1111 1110	14	29	H
2020	0111 1110 0100	4	9	H
257	0001 0000 0001	1	2	M
967	0011 1100 0111	7	14	H
819	0011 0011 0011	3	6	H
1450	0101 1010 1010	10	21	M
958	0011 1011 1110	14	29	M
1092	0100 0100 0100	4	8	M
1928	0111 1000 1000	8	17	M
367	0001 0110 1111	15	31	H
819	0011 0011 0011	3	6	H
455	0001 1100 0111	7	14	M
1010	0011 1111 0010	2	4	M
1023	0011 1111 1111	15	30	M
1092	0100 0100 0100	4	8	H

Cache

Set 0		Set 4		Set 8		Set 12	
Line 0	496 3	Line 8	1092 1	Line 16	504 3	Line 24	684 4
Line 1	240 0	Line 9	2020 3	Line 17	1928 0	Line 25	204 3
Set 1		Set 5		Set 9		Set 13	
Line 2	257 0	Line 10	2005 2	Line 18	1945 1	Line 26	1373 1
Line 3	1937 4	Line 11	1589 4	Line 19	1369 2	Line 27	13 2
Set 2		Set 6		Set 10		Set 14	
Line 4	1010 0	Line 12	102 2	Line 20	1834 4	Line 28	46 3
Line 5	578 3	Line 13	550 1	Line 21	1450 0	Line 29	958 0
Set 3		Set 7		Set 11		Set 15	
Line 6	819 4	Line 14	455 0	Line 22	923 2	Line 30	1023 0
Line 7	339 3	Line 15	2039 2	Line 23	859 2	Line 31	367 4

Number of hits: 11

Number of misses: 9